

$$C_{\text{qud}}^{22,1,-i_2} \rightarrow \frac{1}{16\pi^2}$$

$$\begin{aligned}
& \left(\frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,0} [m_1, m_q^p, m_u^{i_1}] + \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_1, m_q^p, m_u^{i_1}] - \right. \\
& \frac{1}{9} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{3,1,1,-1} [m_1, m_q^p, m_u^{i_1}] + \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,0} [m_1, m_q^p, \tilde{\mu}] - \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{3,1,1,-1} [m_1, m_q^p, \tilde{\mu}] - \\
& \frac{1}{18} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_1, m_u^{i_1}, m_q^p] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_q^p] + \\
& \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_1, \tilde{\mu}, m_u^{i_1}] - \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,0} [m_2, m_q^p, \tilde{\mu}] + \\
& \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{3,1,1,-1} [m_2, m_q^p, \tilde{\mu}] + \frac{3}{2} m_2 \tilde{\mu} g_2^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_2, \tilde{\mu}, m_q^p] + \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,0} [m_3, m_q^p, m_u^{i_1}] + \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_3, m_q^p, m_u^{i_1}] - \\
& \frac{4}{3} m_3 \tilde{\mu} g_3^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{3,1,1,-1} [m_3, m_q^p, m_u^{i_1}] - \frac{2}{3} m_3 \tilde{\mu} g_3^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,2,1,-1} [m_3, m_u^{i_1}, m_q^p] + \\
& \frac{1}{4} \tilde{\mu} \overline{y_d^{p r}} y_d^{p i_2} \overline{y_u^{s i_1}} a_d^{s r} \text{LF}_{2,2,1,-1} [m_d^r, \tilde{\mu}, m_q^s] - \\
& \frac{1}{4} \tilde{\mu} \overline{y_d^{p r}} y_d^{p i_2} \overline{y_u^{s i_1}} a_d^{s r} \text{LF}_{2,2,1,-1} [m_q^s, \tilde{\mu}, m_d^r] - \frac{2}{3} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_1, m_u^{i_1}] + \\
& \frac{2}{3} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_1, m_u^{i_1}] + \frac{1}{2} \tilde{\mu} \overline{y_d^{p r}} y_d^{p i_2} \overline{y_u^{s i_1}} a_d^{s r} \text{LF}_{2,1,1,0} [\tilde{\mu}, m_d^r, m_q^s] - \\
& \frac{1}{2} \tilde{\mu} \overline{y_d^{p r}} y_d^{p i_2} \overline{y_u^{s i_1}} a_d^{s r} \text{LF}_{3,1,1,-1} [\tilde{\mu}, m_d^r, m_q^s] - \\
& \frac{1}{6} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \text{LF}_{1,1,1,1,0} [m_1, m_d^{i_2}, m_q^p, \tilde{\mu}] + \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \\
& \text{LF}_{2,1,1,1,-1} [m_1, m_d^{i_2}, m_q^p, \tilde{\mu}] - \frac{1}{3} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{1,1,1,1,0} [m_1, m_q^p, m_u^{i_1}, \tilde{\mu}] - \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,1,-1} [m_1, m_q^p, m_u^{i_1}, \tilde{\mu}] + \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \\
& \text{LF}_{2,1,1,1,-1} [m_d^{i_2}, m_1, m_q^p, \tilde{\mu}] - \frac{2}{9} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_d^{i_2}, m_u^{i_1}] - \\
& \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_d^{i_2}, \tilde{\mu}] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \\
& \text{LF}_{2,1,1,1,-1} [m_q^p, m_1, m_u^{i_1}, \tilde{\mu}] + \frac{4}{3} \tilde{\mu} g_3^2 \overline{y_u^{p i_1}} a_d^{p i_2} \text{LF}_{2,1,1,1,-1} [m_q^p, m_3, m_d^{i_2}, m_u^{i_1}] - \\
& \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,1,-1} [m_u^{i_1}, m_1, m_q^p, \tilde{\mu}] - \frac{1}{12} \tilde{\mu} g_1^2 \overline{y_u^{p i_1}} a_d^{p i_2} \\
& \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_d^{i_2}, m_q^p] + \frac{1}{6} m_1 \tilde{\mu} g_1^2 y_d^{p i_2} \overline{y_u^{p i_1}} \text{LF}_{2,1,1,1,-1} [\tilde{\mu}, m_1, m_q^p, m_u^{i_1}] \Big)
\end{aligned}$$